SECONDARY STORAGE (STORAGE)

GENERAL VOCABULARY

volatile likely to change, especially quickly or unpredictably

HINT

People often confuse storage and memory. You must know the difference between these terms, including what they do and the ways in which they are used.

SUBJECT VOCABULARY

virtual memory space on a computer for storing instructions and programs until they are needed or being used

motherboard the circuit in a computer with connectors to which other components connect

Secondary storage is often referred to as 'storage'. However, it should not be confused with primary storage or main memory, which are terms sometimes used to refer to Random Access Memory (RAM) or Read Only Memory (ROM). You will learn more about RAM and ROM on pages 59–61. Secondary storage is non-**volatile**, which means it will not lose data when the system's power is turned off.

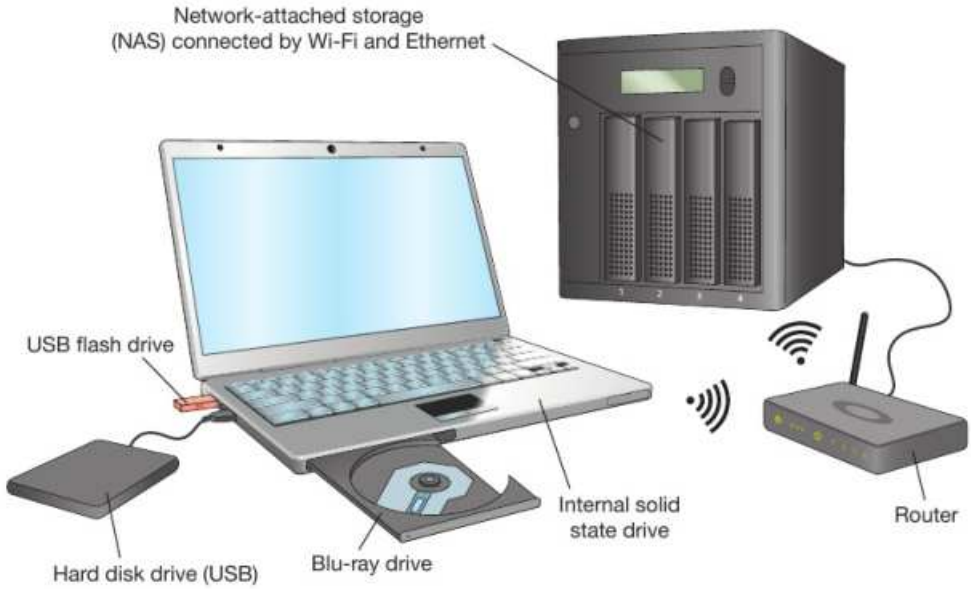
Secondary storage is used for:

- storing a document for future use
- storing an application, ready to be loaded into RAM when the user opens the application
- **virtual memory** (see page 60).

DEVICES

Storage devices are used to store data or software that is used in a computer system.

Storage devices can be either internal or external. Internal drives connect directly to the computer's **motherboard**. External devices are connected to the computer's motherboard through the different ports and adapters on the computer's exterior .



▲ **Figure 1.73** Storage can be internal or external

HINT

You must know the difference between a storage device and storage media. Remember: a storage device *contains* the storage media and *accesses* the data stored on it. For example:

1. flash memory is the media used to store data, whereas a USB drive is used to contain the flash media so that it can be read by the computer after it is plugged into a USB port
2. a DVD disk is the media used to store data, whereas a DVD drive reads the data on the DVD disk.

SUBJECT VOCABULARY

backwards compatibility the ability to be used with older technology without having to be specially adapted

There are different types of storage device.

- Hard disk drives (HDD) contain hard disk media. The drives provide a connection from the disk to the motherboard, either directly or using a wireless adapter like Wi-Fi or a wired port like USB.
- Solid state drives (SSD, often referred to as flash drives) contain flash memory media. They are otherwise identical to hard disk drives.
- Optical disk drives contain optical disk media. Newer drives are often compatible with older media. For example, a disk drive that can read Blu-ray media can also read DVD and CD media. This is known as **backwards compatibility**.

GENERAL VOCABULARY

concentric (of circles or circular objects) sharing the same central point
spindle a rod around which something spins

SUBJECT VOCABULARY

platters circular plates
read/write head the part of a disk drive that passes across (or floats) the platters on a very thin layer of gas above the platters
seek time the time it takes for a read/write head to locate the area on the disk where the data to be read is stored

DID YOU KNOW?

The gap between the platter and the read/write head is only a few nanometers wide. This means that a dust particle or even a smoke particle could cause a collision between the platter and the read/write head. To prevent this, hard drives use filters in their ventilation systems or are built as sealed units.

HINT

If a read/write head crashes into a platter, it can cause damage to data and the platter. This would not happen with a solid state drive.

DID YOU KNOW?

Optical media layers deteriorate (get worse) over time, so they have to be stored carefully to prevent data loss.

DID YOU KNOW?

Blu-ray disks get their name from the colour of laser used. The laser light is actually a violet colour, but perhaps 'violet-ray' was not considered a good enough name.

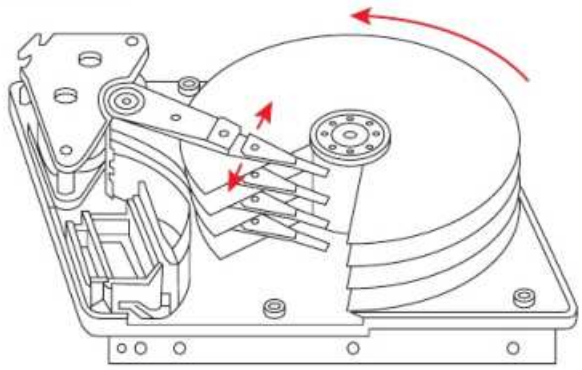
MEDIA

Hard disks

Hard disks are made up of many **concentric platters**. These platters make up a cylinder that spins on a central **spindle**. A **read/write head** moves on an arm across tracks on the platter. The amount of time that it takes the read/write head to access data on the tracks is determined by how fast the cylinder of platters spins and how fast the read/write head is moved across the tracks. When reading, a read/write head changes the magnetic field into electrical current. When writing, it transforms electrical current into a magnetic field.

Typical spin speeds are 5400 revolutions per minute (rpm) or 7200 rpm.

The average **seek time** for a read/write head is 4–15 milliseconds (ms).



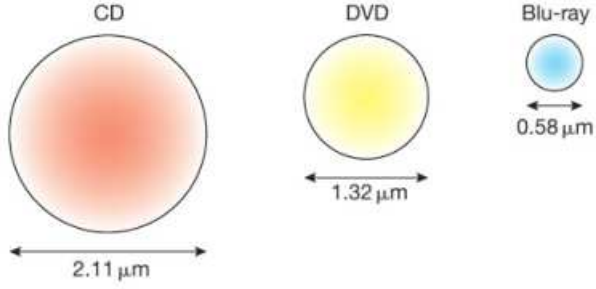
▲ **Figure 1.74** Read/write heads move across a hard disk platter to access data as the cylinder spins

Optical media

Types of optical media include CDs, DVDs and Blu-ray disks.

- **CDs** can store up to 700 MB of data. The data is written to the disk using a laser, which writes data to a plastic layer beneath layers of aluminium and acrylic.
- **DVDs** look very much like CDs. Data is written to DVDs using a shorter wavelength of red laser light, which allows DVDs to store more data. They can store 4.7 GB on a single-sided, single-layer disk. Double-sided, dual-layer disks can store as much as 18 GB.
- **Blu-ray** disks are similar to CDs and DVDs, but use a scratch-protection coating. Violet laser light is used to store data at greater density than the red laser light used in DVDs. They can store 25 GB on a single-sided disk, and double-sided disks can store 50 GB.

All optical media is recordable and is known as CD-R, DVD-R or BD-R. Some CDs and DVDs can be rewritten (CD-RW, DVD-RW), and some Blu-ray media can be written then erased and rewritten (BD-RE).



▲ **Figure 1.75** Comparison of various lasers used to store data on optical media

DID YOU KNOW?

One explanation of why flash memory is so-called is that each cell is flashed (flooded) with electrons, forcing it to hold its charge. One other explanation for the name is that data is written very quickly (or, in colloquial English, 'in a flash').

Flash media

Flash media are more energy-efficient than hard disks as they do not have moving parts. For the same reason, they are also less likely to fail when they are moved around. This makes them suitable for use in portable devices.

Magnetic tape

Magnetic tape was originally designed to record sound, but it is now also used to store data.



▲ **Figure 1.76** Magnetic tape is mainly used for backing up data in large organisations

SUBJECT VOCABULARY

data access speed how quickly data can be read from or written to media
capacity the amount of data that can be stored on media
bit (binary digit) a single unit of information with a value of either 0 or 1; eight bits equal one byte

▼ **Table 1.4** Comparing types of media

	HARD DISK	FLASH MEDIA	OPTICAL MEDIA	MAGNETIC TAPE
DATA ACCESS SPEED	Fast	Fastest	Slow	Slowest
MAXIMUM CAPACITY	Varies, up to 128 GB	Varies, up to many TB	CD 700 MB DVD 18 GB Blu-ray 50 GB	Varies, up to 185 TB
COST PER GB	High	Very high	Medium	Low
USE	Servers, personal computers, backups	Laptops, mobile devices	Multimedia (music, games and films), file backups	Whole system backups and archives
PORTABILITY	Not suitable	Yes	Not suitable	Not suitable

Storage media store data in binary form. This means that each **bit** of data holds one of two values: 0 or 1. Different media types interpret 0 or 1 differently.

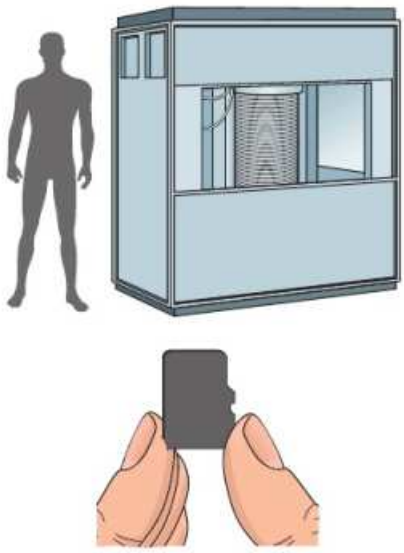
- Hard disks change the magnetic charge of a platter to either negative or positive, depending on whether the value is 0 or 1.
- Optical media use tiny bumps on the disk's surface to represent 0 and 1.
- Flash media use different levels of electrical charge, held in tiny individual cells, to represent 0 or 1.
- Magnetic tape changes the magnetic charge of the tape to either negative or positive, depending on whether the value is 0 or 1.

DID YOU KNOW?

The 'bi' in '**bi**nary' comes from the Latin for 'two'.

HINT

You do not need to understand the units shaded in dark grey in the table for the International GCSE.



▲ **Figure 1.77** The IBM 350 in comparison with a person and a 256 GB microSD card

▼ **Table 1.5** Units of storage capacity

NUMBER OF BYTES	WITH UNIT SYMBOL	NAME
1000	1 KB	Kilobyte
1000,000	1 MB	Megabyte
1000,000,000	1 GB	Gigabyte
1000,000,000,000	1 TB	Terabyte
1000,000,000,000,000	1 PB	Petabyte
1000,000,000,000,000,000	1 EB	Exabyte
1000,000,000,000,000,000,000	1 ZB	Zettabyte
1000,000,000,000,000,000,000,000	1 YB	Yottabyte

The IBM 350 was the first disk drive. It was 172 cm high, 152 cm long and 74 cm wide, and it could store 3.75 MB of data. People could hire one for US$38,400 per year. In 2016, manufacturers started selling the 256 GB microSD card, which is a flash memory card the size of a human fingertip. It was priced at US$175 and can store the same amount of data as 68,266 IBM 350 disks!

DID YOU KNOW?

In 2017, it would take about 9457 trillion years to download one yottabyte of data with a good internet connection.

CHAPTER QUESTIONS

SKILLS CRITICAL THINKING

1 State **one** benefit of using barcodes in a supermarket. **(1)**

SKILLS REASONING

2 State **one** reason why small, built-in microphones are used in smartphones and laptops. **(1)**

SKILLS DECISION MAKING

3 Explain why capacitive touch screens do not work when you are wearing woollen gloves. **(2)**

4 State **three** computer systems that **do not** use monitors. **(3)**

SKILLS CRITICAL THINKING

5 State **two** benefits of using laser printers rather than inkjet printers. **(2)**

SKILLS PROBLEM SOLVING

6 State **two** types of business that use plotters. **(2)**

SKILLS REASONING

7 Explain why some 3D printers use biodegradable materials to create models. **(3)**

SKILLS CRITICAL THINKING

8 List **two** sensors used in smartphones. For each one, describe one way in which it is used by the operating system or an app. **(4)**

SKILLS REASONING/ARGUMENTATION

9 Discuss the advantages and disadvantages of using sensors in smart cars. **(8)**

SKILLS DECISION MAKING

10 Explain **one** reason why Blu-ray disks are used to store modern console games. **(2)**

2 SOFTWARE

The programs that run on a computer are called software. Programs are instructions that are carried out or executed by the computer's processor, which provides other instructions for the rest of the computer.

Some computers can only run one program at a time, while others can run multiple programs at the same time. Some computers are single purpose, which means that they are only able to run one piece of software (for example, the software used to control a microwave). Other computers are general purpose, which means that they are able to carry out a range of tasks (for example, the software used in a tablet computer).

There are two basic groups of software: system software and application software.



LEARNING OBJECTIVES

- Be able to identify the purpose of systems software and applications software
- Know about operating systems and system software tools such as utilities
- Know about the role/function of the operating system, including basic knowledge of:
  - single user and network
  - memory management
  - resource management
  - security
  - print spooling
- Know about software applications (apps), including office-productivity tools, web authoring, image and sound editing, presentation software, control software, project management software
- Know about software licensing types (free/open source and proprietary software)
- Understand that the purpose of communication software is to provide remote access to systems and to exchange files and messages in text, images, audio and/or video formats between different computers or users
- Know why software is updated, how it is done and possible risks to data/systems
- Understand that settings of ICT systems can be configured to meet the accessibility needs of individuals
- Be able to justify choices made in identifying and configuring hardware and software

SYSTEM SOFTWARE

Programs that are designed to maintain or operate the computer system are known as system software.

UTILITY SOFTWARE

Utility software is one form of system software, which carries out configuration and maintenance tasks.

SUBJECT VOCABULARY

utility software (also known as utilities) system software that carries out configuration and maintenance tasks

BACKUP

Backup utilities create a copy of files and programs. Backups can be set to run automatically (usually at a time when the system is not in use) or can be started by a user.

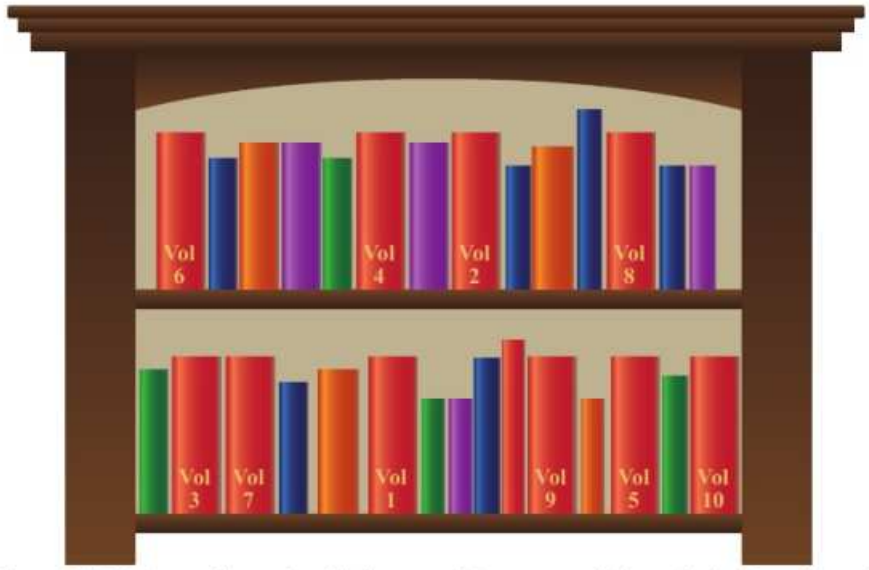
You will learn more about backups in *Unit 3 Operating online* (pages 104–105).

DEFRAGMENTATION

GENERAL VOCABULARY

fragmented broken into pieces

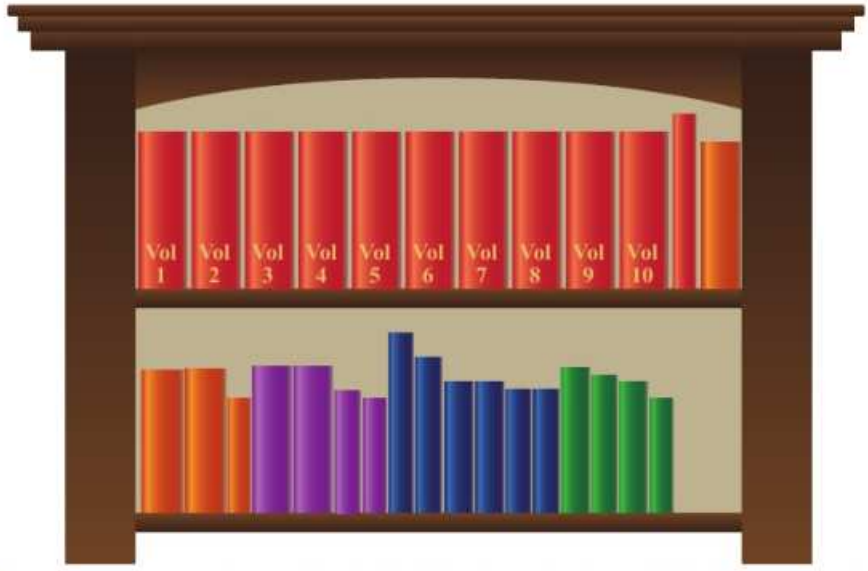
As data is stored to hard disk, some systems spread it across the disk wherever there is free space. This means that sometimes data is **fragmented** and stored out of order or out of sequence.



▲ **Figure 2.1** Data is sometimes stored in the nearest free space, which results in data getting out of sequence. This is an illustrative example.

Data from the same file may be spread over multiple locations on a platter, on different platters in the cylinder or sometimes even on different disks. When these files are loaded, data has to be retrieved from these multiple locations. The read/write head of the hard drive has to find this data, which takes extra time and slows down the process of loading the files.

To avoid this and to speed up the system, disk defragmentation utilities reorder the fragments of data so that they are stored as close to each other as possible.



▲ **Figure 2.2** Defragmentation puts the data back in order so that the read/write head does not have to search the disk for fragments of data. This is an illustrative example.

Defragmentation utilities are usually set to run automatically, but they can also be started by a user.

SKILLS EXECUTIVE FUNCTION
INTELLECTUAL INTEREST AND CURIOSITY
SELF-DIRECTION

ACTIVITY

▼ DEFRAGMENTATION

1 Run a defragmenter analysis on your system. When was the analysis last run? How fragmented was the data last time the analysis ran?

2 Research the need for defragmentation on SSDs.

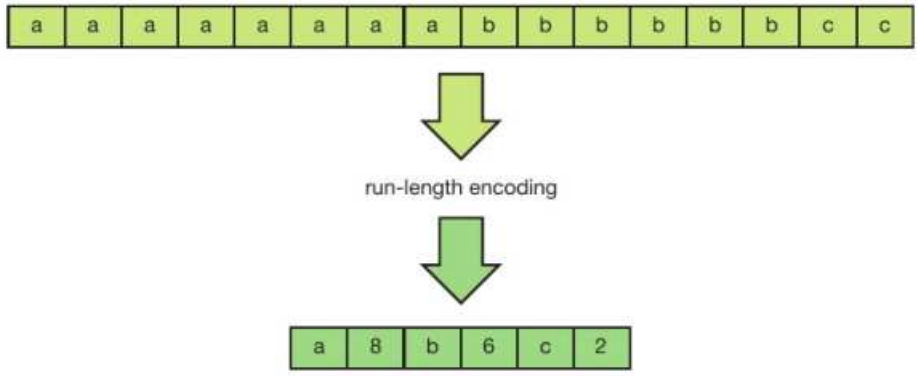
COMPRESSION



▲ **Figure 2.3** Data can be compressed to remove unnecessary use of code or space in the file. This is an illustrative example.

Compression utilities reduce the original size of a file or set of files.

Where there are repeated patterns of data, rather than storing every repeated instance, only the first instance of the data is stored, alongside how many times it is repeated. For example, in Figure 2.4, there were eight instances of a, six instances of b and two instances of c in the original file. This data can be compressed to be stored as 'a8b6c2', which reduces the amount of storage needed.



▲ **Figure 2.4** Run Length Encoding is one method of data compression

SUBJECT VOCABULARY

decompress extract data from a compressed file so that it can be read

The resulting compressed file is not usually readable by the original application. Compression utilities also have the ability to **decompress** (extract) the data from a compressed file so it can be read by the original application again.

FORMATTING

Disk formatting prepares storage media such as a hard disk drive or USB flash drive for its first use. If a disk has already been used, then formatting it will make all of the data on the disk unreadable by normal applications.

DID YOU KNOW?

Different operating systems require disks to be formatted in certain ways in order to perform specialist tasks. Disks can also be formatted in a way that makes them compatible with all operating systems.

Disk formatting utilities make this process easy for users by providing them with the option to choose the file system, file unit sizes and the name of the disk. Perhaps most importantly, they make sure that the user really wants to format the disk.

14 FILE MANAGEMENT

Over your lifetime, you will probably create many thousands of digital files. Unless you are organised, these files can easily get lost on your devices. Managing your files carefully will save you time and will help you to find the files that you are working on. It will also help you to locate old files that you can repurpose for a different use.

Naming your files properly will ensure that you always work on the correct version of a file. It is also important to use the correct file types, because this allows you to share your files with other people and guarantees that they will be able to view the finished product.

File security is another important part of file management. This will protect your files and prevent unauthorised users from changing their contents.



LEARNING OBJECTIVES

- Save work regularly and keep information secure
- Use sensible filenames and formats
- Create and manage files and folder structures

SAVING WORK

When you are creating digital products, it is important that you save your work regularly. This is to make sure that you have a copy of your work in case the application crashes or you lose power to your computer. This should include saving your work frequently to secondary storage. For more information about secondary storage, see *Unit 1 Digital devices* (pages 38–41).

SKILLS EXECUTIVE FUNCTION
SELF-DIRECTION
REASONING

ACTIVITY

▼ SAVING A FILE

1. Create a new word processor document and save the document with the filename 'time'.
2. Time how long it takes you to save a file using each of the following methods.
   - Use a keyboard shortcut to save the file to secondary storage.
   - Use the mouse to save the file to secondary storage.
3. Compare the time taken to save the file by each method. Which one took longer?
4. Discuss why you think different methods of saving a file are available.

HINT

The keyboard shortcut to save a file is Ctrl+S on a Windows computer or Command+S on OS X for Apple Mac®.

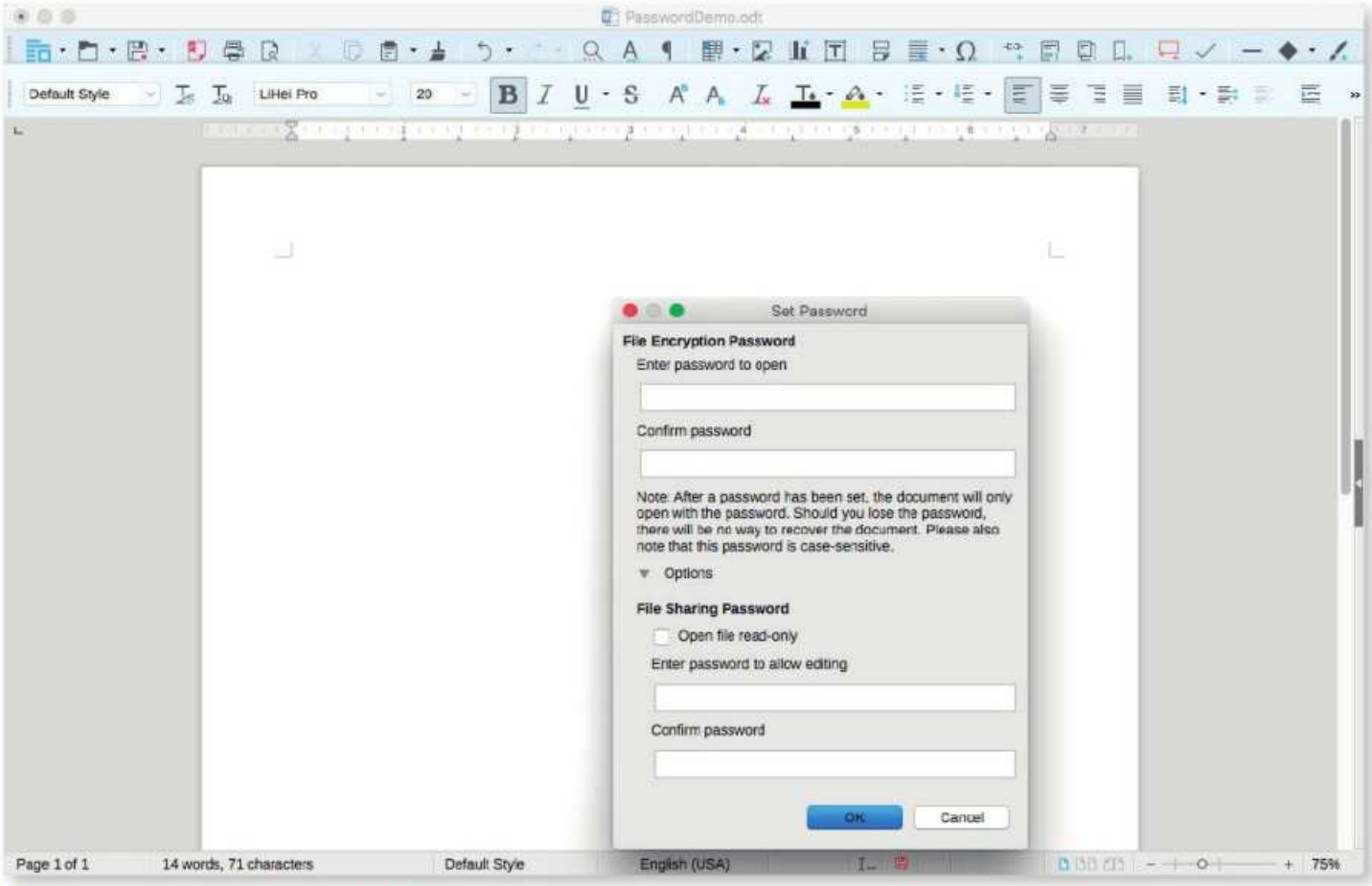
DID YOU KNOW?

Many applications will save your work to a temporary file. You may be able to use this to recover your work if the application crashes.

SECURING FILES

You can secure your files by adding a password to be able to read or edit the file, as shown in Figure 14.1.

For more information about passwords and encryption, see *Unit 3 Operating online* (page 100) and *Unit 2 Connectivity* (page 90)



▲ **Figure 14.1** Applications often have **built-in** features that allow you to secure your files

SKILLS EXECUTIVE FUNCTION

GENERAL VOCABULARY

built-in forming a part of something, from which it cannot be separated

HINT

If you are using Microsoft® Word, you can protect the document with a password using the Protect Document options in the Info section of the File menu.

ACTIVITY

▼ SECURING A FILE

1. Create a new word processor document and type a secret message into it.
2. Protect the document with a simple password that you can remember easily.
3. Save the document and close it.
4. Open the document again. What happens?

NAMING FILES

It is important to use sensible filenames. A sensible filename describes the contents of a file so that it can be found easily.

For example, imagine that you are creating a poster for a sports club and have to name your file. The following points are some of the considerations that will help you to decide on a good filename.

- You might choose to name it Poster. However, you might make more posters later.
- You could name it First Poster, to be followed by Second Poster, Third poster and so on, but in the future you may not remember what Second Poster referred to.
- You could name it Sports Club Poster, but what would happen if you were asked to create a different sports club poster in the future? What if you needed to create more than one poster for the same club?

It is often a good idea to add the date of the event or the file creation date in the filename. When you add a date to a filename, you should consider using reverse date order, which means giving the date in the order: year, month, day. For example, an invitation to a party on the 26th December 2019 could be dated 191226. This means that, if you create lots of different party invitations throughout the year, you can sort them by name order so that they appear in the order of the date of the parties. Because the date is reversed, the filenames are sorted by year first, then month, then day. This is not possible if you write dates in the conventional order: day, month, year.

SUBJECT VOCABULARY

snake case using underscores (underlines) to separate words in a filename

If you need to use more than one word in your filename, you should not use spaces between the words. This is because the spaces can cause problems with some software. Instead, you can use **snake case**. Snake case is useful because putting_an_underscore_between_each_word is easier to read than puttinganunderscorebetweeneachword.

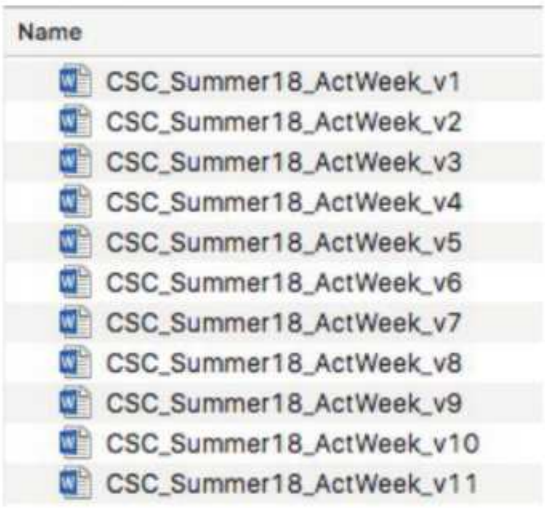
Table 14.1 shows and explains two sensible filenames.

▼ **Table 14.1** Examples of good filenames

PRODUCT	FILENAME	REASON
A poster for the City Sports Club's Summer 2018 Activity Week	CSC_Summer18_ ActWeek	• CSC for City Sports Club • Summer18 for the date • ActWeek for the event (Activity Week) • _ used to separate words (snake case)
A presentation to your school about a fundraising day on 16th November 2018	School_Charity_ Presentation_181116	• Describes the event • Uses snake case • Gives the date in reverse so that the files will appear in date order when they are sorted by name

VERSIONING

As you make changes to a file, you can name it with a version number. This is known as versioning and it allows you to return to earlier versions of the same document at a later date. For example, the first version of CSC_Summer18_ ActWeek from Table 14.1 could be called CSC_Summer18_ActWeek_v1, as shown in Figure 14.2.



▲ **Figure 14.2** An example of versioning

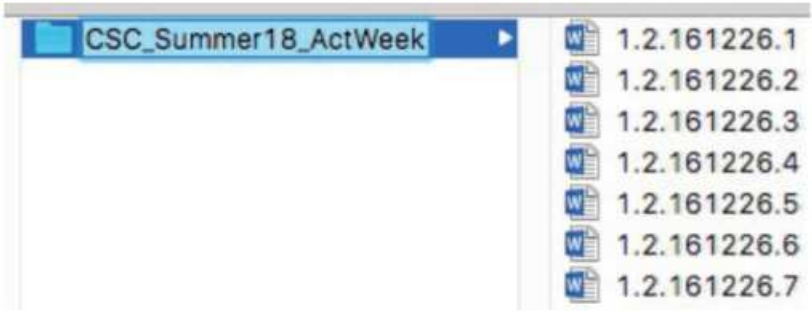
You could also include the date in your versioning by using the format MAJOR.REVISION.BUILDdate.BUILDnumber. For example, 1.2.161226.5 specifies the following information.

SUBJECT VOCABULARY

build date the date on which a file was created
build number the number within a series of versions from the build date

- MAJOR: version 1
- REVISION: 2
- **BUILDdate**: 161226 (26th December 2016)
- **BUILDnumber** (from that BUILDdate): 5

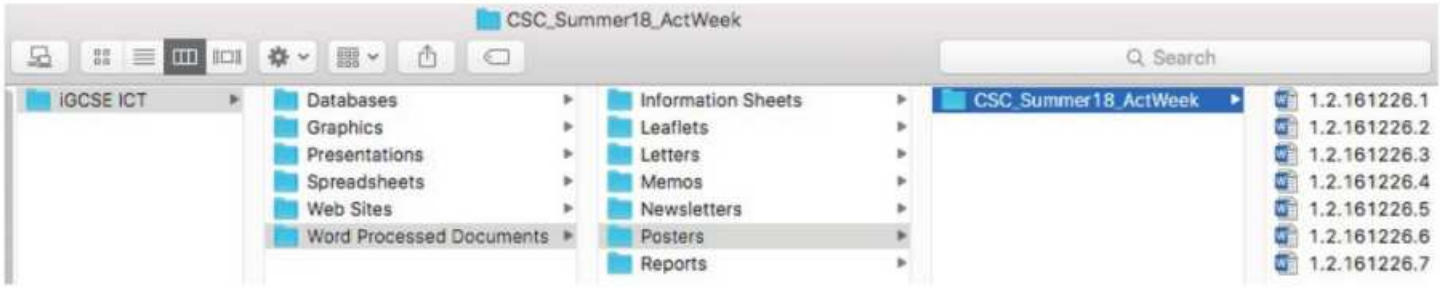
The filename is now CSC_Summer18_ActWeek_v1.2.161226.5. However, this filename is very long. To reduce its length, you could organise your files in a folder. If you create a folder to hold all the versions of the CSC Summer 2018 Activity Week files, as shown in Figure 14.3, you don't need to repeat the same text in the filenames.



▲ **Figure 14.3** A folder titled CSC_Summer18_ActWeek containing versions of files

FOLDERS

You can use a folder structure to keep your folders and files organised. For example, in Figure 14.3, none of the files used the word 'poster' in their filename. This means that you may not be able to tell that they were posters. However, because the CSC_Summer18_ActWeek folder contains poster files, you could place it inside a folder called Posters. This ensures that your files remain organised and also lets you know what type of documents are inside the folder. In turn, the Posters folder could be placed inside a folder called Word Processed Documents, as shown in Figure 14.4.



▲ **Figure 14.4** Keep your folders organised so that you can find your files easily

SKILLS EXECUTIVE FUNCTION

ACTIVITY

▼ CREATING A FOLDER STRUCTURE

Create folders for your International GCSE ICT work so that you can keep all your files organised.

FILE FORMATS

GENERAL VOCABULARY

compatible two or more things that are able to function properly when they are used together

Software applications can only read files that are saved in **compatible** formats. For example, presentation applications cannot open files saved in a format used by spreadsheet applications. It is good practice to save your files in a format that will be compatible with the software that you will use to edit it.

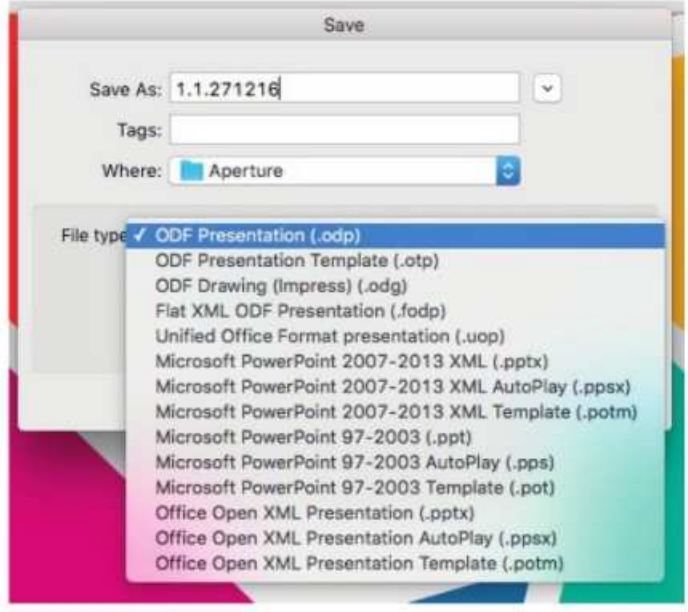
When saving files, you can choose to save them in different formats. For example, you might choose to save your work in an older format so that it can be read by people who use older versions of the software. Software is often backwards compatible, which means that newer versions of application software can read files created using older versions. However, sometimes you will find that older versions of the software cannot read files created using a newer version.

HINT

Formats are often identified by **file extensions**.

SUBJECT VOCABULARY

file extension the letters at the end of the name of a computer file, which show what type of file it is, such as .doc



▲ **Figure 14.5** Files can be saved in different formats, such as ODF Presentation (.odp) in this example

Files that are saved in application file formats are often only able to be viewed and edited by applications that are compatible with that format. See Table 14.2 for examples.

DID YOU KNOW?

Published file formats can usually be viewed in a web browser, whereas application file formats usually cannot be viewed in a web browser.

KEY POINT

You need to use specific application software in order to open any files stored in that application's file format. In comparison, published file formats can be read by a wider range of applications.

When you have finished editing a file, you may wish to publish or export it. Publishing or exporting the contents of the file in a published file format creates a final version of the file. Your audience will be able to view the file, but they will not be able to change it easily using application software.

SUBJECT VOCABULARY

transparency areas of an image that do not have a colour value, including white, are made see-through; if transparency is unavailable due to the file format, those areas are displayed as white

▼ Table 14.2 Examples of application file formats and their related published file formats

APPLICATION TYPE	APPLICATION	APPLICATION FILE FORMAT	PUBLISHED OR EXPORTED FILE FORMAT
Word processor	Microsoft® Word	.docx	.pdf
Presentation	LibreOffice® Impress	.odp	.ppsx
Graphics	GIMP	.xcf	.jpeg .png
Web authoring	Serif® WebPlus	.wpp	.html

Different published graphics file formats have different properties. For example, to keep the **transparency** of an image, you could choose to export it as a .png file.

CHAPTER QUESTIONS

SKILLS PROBLEM SOLVING

1 Which **one** of these is the most sensible filename for a logo created for a school? **(1)**

A Logo-1

B FinalLogo

C School_Logo_v1.02

D 102 School

SKILLS REASONING

2 Explain why it is important to save work regularly. **(3)**

SKILLS REASONING

3 State the reason why including versions in filenames is useful when developing a digital product. **(1)**

SKILLS PROBLEM SOLVING

4 State **three** published file formats. **(3)**

SKILLS INTERPRETATION

5 Describe how a user could use the MAJOR.REVISION.BUILDdate. BUILDnumber filename format to help them to find files at a later date. **(3)**

SKILLS PROBLEM SOLVING

6 State **one** method used to secure a file. **(1)**

SKILLS REASONING

7 State the reason why exporting a file is different from saving the file. **(1)**